

2.3.3 Number of Repetitions

دہرائی (لوپ) کارکردگی کا سب سے بڑا عامل ہے۔

CS XII

Chapter 2: Computational Thinking & Algorithms

Lecture 08 of 16

2.3.3 Number of Repetitions

LEARN IT

Repetitions (loops) are the most significant factor in algorithm efficiency. The **structure of the loop** decides how time grows with n .

Type of repetition	Conceptual impact	Efficiency grade
Linear loop	Processes each element exactly once. Time grows in proportion to size.	Good — $O(n)$
Logarithmic repetition	Reduces the problem size by half each iteration (binary search). Huge time savings.	Excellent — $O(\log n)$
Nested loops	For each element, process all other elements. Operations explode.	Poor — $O(n^2)$

WORKED EXAMPLE

Worked example: count the inner-body runs.

```
n = 5
linear = 0
for i in range(n):
    linear = linear + 1

nested = 0
for i in range(n):
    for j in range(n):
        nested = nested + 1

print(linear, nested)
```

OUTPUT
5 25

One loop: 5. Nested loops: 25 = 5^2 . For $n = 1,000$ that would be 1,000 versus 1,000,000.

REMEMBER IT

Find the **deepest loop**. One loop $\approx O(n)$. Loop inside a loop $\approx O(n^2)$. A loop that halves $n \approx O(\log n)$.

COMMON MISTAKE

Mistake: counting the word `for` and ignoring how the index moves.

Fix: a loop `while n > 1: n = n // 2` is logarithmic, even though it is still “a loop”.

دہرائی (لوپ) کارکردگی کا سب سے بڑا عامل ہے۔ ایک بار ہر عنصر پر چلنے والا لوپ $O(n)$ (اچھا)، ہر بار مسئلہ آدھا کرنے والا $O(\log n)$ (بہترین)، اور اندرونی لوپ $O(n^2)$ (خراب) ہے۔ اندرونی جسم کے چلنے کی گنتی سے یہ فرق صاف نظر آتا ہے۔

CHECK YOURSELF

1. What grade does the source give nested loops?
2. Why is binary search's loop logarithmic rather than linear?
3. If $n = 5$, how many times does a fully nested n -by- n pair of loops run?

Answers: 1. Poor — $O(n^2)$. 2. Each iteration halves the remaining problem, so the number of iterations is about $\log n$. 3. 25.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. for i in range(3) runs:

- a. i=0,1,2
- b. i=1,2,3
- c. i=3,2,1
- d. forever

MCQ 2. while repeats while the condition is:

- a. True
- b. False
- c. A file
- d. None always

MCQ 3. break:

- a. Exits the loop immediately
- b. Skips one line only
- c. Closes Python
- d. Commits SQL

MCQ 4. continue:

- a. Skips the rest of this iteration
- b. Ends the program
- c. Defines a function
- d. Opens a file

MCQ 5. A nested loop: inner runs:

- a. Fully for each outer step
- b. Once in total
- c. Never
- d. Only if NAND

MCQ 6. range(2, 6) yields:

- a. 2,3,4,5
- b. 2,3,4,5,6
- c. 1,2,3,4,5,6
- d. 6,5,4

Extra short questions

1. When for vs while?
2. What is an infinite loop?
3. Trace s=0; for i in range(1,4): s+=i.
4. Shopping bill uses which loop typically?

Extra long question

1. Write nested loops to print a 3×3 table of i,j. Then rewrite the shopping-bill outline with a while, a running total, and an if on quantity.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. for: known count / iterate a sequence. while: unknown count, stop on a condition.
2. while True without break, or a condition that never becomes False.
3. i=1 s=1; i=2 s=3; i=3 s=6.
4. while more items (unknown count) or for a known list of items.

Long

1. for i in range(3): for j in range(3): print(i,j). Bill: total=0; while True: read item/qty; if qty<=0: continue; if item=="end": break; total+=...; print total.